

1.

1.1.

(22) **Rural farmers** – farmers have a frustrating, stressful, and discouraging UX. This is because they have limited access to the market, poor transportation, bad roads and no platform to connect them with buyers or transport providers and they end up losing income and wasting food.

Traders (markets, hawkers, spaza shops) – they experience ^{not UX} inefficiencies and inconvenience when sourcing fresh produce. This is because of no central system showing farmers with available produce. They then spend lots of time searching and relying on word of mouth

Households – they experience uncertainty, anxiety, and food security. They do not know where their next meal will come from. Although there is a surplus of food, there is no communication system which connects them with farmers. This leaves them hungry while food is wasted

Transport providers – transport providers have an uncoordinated and underutilised UX. There is no platform matching them with farmers who may need their services. As a result, transport resources are not used

1.2

Effectiveness - The current process is not effective for farmers because they are unable to successfully get their surplus produce to buyers or households

Efficiency - The efficiency goal is not met because there is no organised way to distribute food, causing delays, wasted time, and difficulty in finding buyers or transport

Utility - The utility goal is not met because there is no system that provides the necessary functions for farmers to advertise surplus produce or for traders and transport providers to coordinate collection and deliver

Learnability: This goal is not met because there is no structured platform or process that users can easily learn to use when sharing, requesting or transporting surplus produce.

Safety: This goal is partially met. Although the scenario does not describe direct physical safety risks, surplus produce is left to spoil in the fields, which may affect food quality and reduce the safety of food intended for consumption

Memorability: This goal is not met because the current process relies on informal communication instead of a consistent system. Users must repeatedly work out how to find buyers, transport or available food, making the process difficult to remember and repeat.

Overall, the current situation does not meet most of the usability goals. This demonstrates the need for a user-centred solution that improves the effectiveness, efficiency, utility, learnability and memorability of distributing surplus food.

1.3

Primary users:

- **Small-scale rural farmers** – They will use the system to register surplus produce, making it easier to find buyers, food relief organisations or households in need. This helps reduce food waste and increases income opportunities.
- **Traders (hawkers, market vendors and spaza shop owners)** – They will use the system to search for available produce from nearby farmers, enabling them to obtain fresh products quickly and reliably.
- **Transport providers** – They will use the system to view and accept transport requests, helping to move produce from farms to traders or vulnerable households.
- **Vulnerable households** – They will use the system to locate available food and request assistance, improving access to affordable or donated produce.

Stakeholders:

- **Local municipalities** – They have an interest in reducing food waste, improving local food security and supporting community development.
- **Government departments such as Agriculture, Transport and Social Development** – They may provide policies, funding and support to ensure the success of the system.
- **Non-governmental organisations (NGOs) and food relief organisations** – They can use the system to identify communities in need and coordinate food distribution programmes.
- **Agricultural extension officers and community leaders** – They can assist farmers with using the system, encourage participation and help coordinate activities within local communities.

Who would provide the solution? Carefully read the last paragraph of the scenario again.

1.4

A potential solution is a central digital platform that aims to address the communication and coordination challenges between the different stakeholders, rather than assuming that the main problem is a lack of food production. The platform could enable farmers to find buyers for their surplus produce, traders and households to find available food, and transport providers to coordinate the collection and delivery of produce. However, before deciding on a digital platform, research would be conducted to determine whether it is accessible and appropriate for the intended users, considering factors such as network availability, smartphone access, data costs and digital literacy.

1.5

Visibility: Visibility ensures that the important features of the system are easy to find and understand. Farmers should be able to clearly see options for adding surplus produce, traders should easily find available products, and transport providers should quickly identify

delivery requests. A clear and well-organised interface will reduce confusion and help users complete tasks successfully, even if they have limited digital experience. ✓

Feedback: Feedback ensures that users receive immediate confirmation after performing an action. For example, the system should notify farmers when their produce has been successfully listed, inform transport providers when a delivery request has been accepted, and notify traders or households when produce becomes available. Providing clear feedback reassures users that their actions have been completed successfully and improves confidence in using the system. ✓

2

a. Lesson one: Finding new inroads to wicked problems

b. I would apply this lesson by first investigating the food insecurity problem from different perspectives rather than immediately deciding that a digital platform is the answer. I would conduct interviews and observations with rural farmers, vulnerable households, traders and transport providers to understand how food is currently produced, sold, transported and accessed. I would investigate issues such as poor roads, transport costs, communication methods, access to technology, affordability and how surplus produce is currently handled. I would also look for relationships between these challenges to understand the wider problem rather than focusing on only one stakeholder. The information gathered would help the design team reframe the problem and identify opportunities that may not initially be obvious. This could reveal that the real need is not simply to create a marketplace, but to improve coordination between food producers, buyers, households and transport providers.

[147 words]

How would you build a trusting relationship that will allow you to talk to users / stakeholders?

-2